2.6.1) Show that the convolution of a tempered function with a Schwartz function is another tempered function.

Proof: Let $f: \mathbb{R}^n \to \mathbb{C}$ be tempered and let $g \in \mathcal{S}(\mathbb{R}^n)$.

Then, there exists M, C, such that $|f(x)| \leq C(1+|x|)^M$, for all $x \in \mathbb{R}^n$. $|f * g(x)| = \left| \int_{\mathbb{R}^n} f(x-y) g(y) \, dy \right|$

$$\leq \int_{\mathbb{R}^n} |f(x-y)| \, |g(y)| \, dy.$$

$$\leq \int_{\mathbb{R}^n} C(1+|x-y|)^M \, |g(y)| \, dy.$$

$$= \int_{|y| \leq \frac{|x|}{2}} C((1+|x-y|)^M |g(y)| \, dy + \int_{|y| \geq \frac{|x|}{2}} ((1+|x-y|)^M |g(y)| \, dy.$$

$$\leq \int_{|y| \leq \frac{|x|}{2}} C\left(1+\tfrac{3|x|}{2}\right)^M |g(y)| \, dy + \int_{|y| \geq \frac{|x|}{2}} ((1+|x|+|y|)^M |g(y)| \, dy.$$

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$$\leq C\left(1+\frac{3|x|}{2}\right)^N \int_{|y|\leq \frac{|x|}{2}} |g(y)|\,dy + \int_{|y|\geq \frac{|x|}{2}} \left((1+2|y|+|y|\right)^N |g(y)|\,dy$$

$$= \left((1+\frac{3|x|}{2}\right)^N \int_{|y|\leq \frac{|x|}{2}} |g(y)|\,dy + \int_{|y|\geq \frac{|x|}{2}} \left((1+3|y|\right)^N |g(y)|\,dy.$$

Notice that by definition, $\left((1+3|y|\right)^N |g(y)|$ gives another Schwartz function, call it φ. So

$$|f*g(x)| \leq C\left(1+\frac{3|x|}{2}\right)^N \int_{|y|\leq \frac{|x|}{2}} |g(y)|\,dy + \int_{|y|\geq \frac{|x|}{2}} \varphi(y)\,dy.$$

Clearly, because any Schwartz function is controlled by $(1+|x|)^{-N-1}$, it is integrable. So,

$|f*g(x)| \leq C\left(1+\frac{3|x|}{2}\right)^N \cdot C_1 + C_2.$

$\leq C_3(1+|x|)^N$. Thus, $f*g$ is tempered. □

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